

Privacy-preserving methods: multimedia and biometrics perspectives

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<http://sip.unige.ch/education/multimedia-security/>

Lecture Outline

- Introduction: privacy and security in biometrics and multimedia
- Cryptographic approach
- Signal processing approach
- Hybrid approach

Information security

- Personal Identifier Number (PIN): to control access to service. svstems and information



- Subscriber Identity Module (SIM): access to mobile phones, to protect against illegal use of network



- Secure Socket Layer (SSL): to prevent eavesdropping on information exchanged between two terminals



Traditional cryptography

Cryptography: develops tools for protecting information and establishing secure interactions.

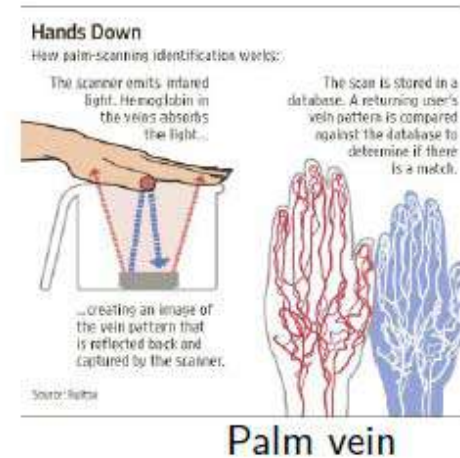
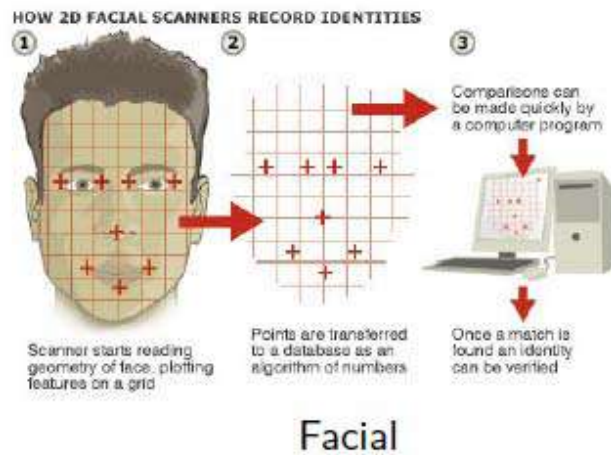
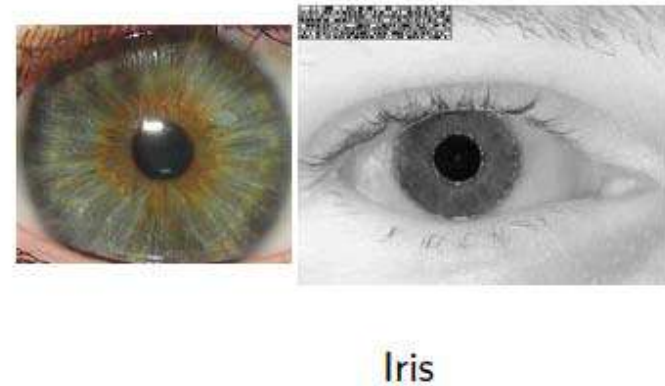
- Encryption: provides confidentiality of information
- Authentication: verifies the identity

Note: The algorithms are public, therefore, the security of the schemes depends on the secrecy of a key owned by the legitimate parties.

Practical cryptography issues in our modern society:

- 1 Key generation and distribution
- 2 Safely storage: (mobile phones, bank tokens, smart cards, : : :) 
- 3 Noisy observation: authentication based on something closely linked to physiological characters 


Biometrics

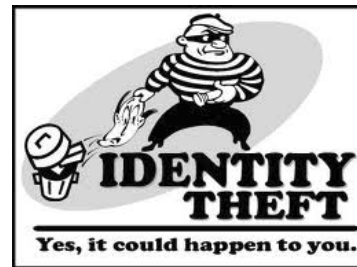


Biometrics: why to be protected?

A biometric template is a representation of a unique characteristic of an individual. It might also contain some information about diseases. Therefore, it contains privacy-sensitive information.

To abuse database:

- impersonation
- identity theft
- cross-database matching
- detectable pathologies
- ... yet undiscovered attacks

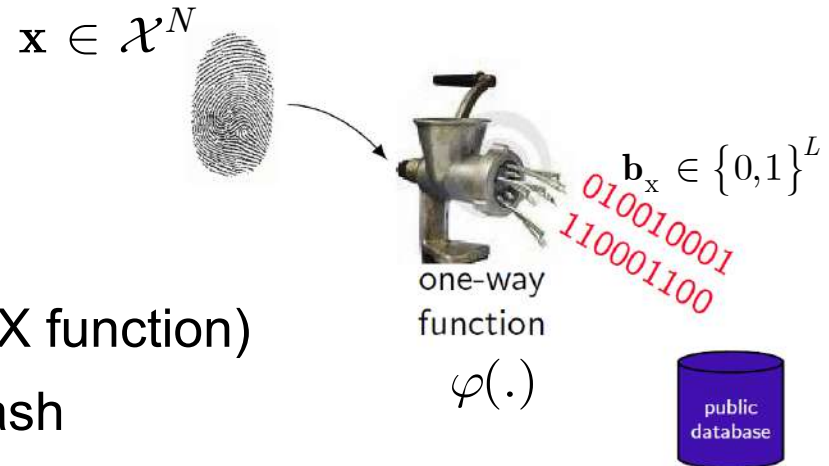


Note: biometric templates can not be "cancelable" because people have only ten fingers, two eyes, etc.

Multimedia security: cryptographic solution

■ Main idea:

- do not store data itself
- store a one-way hash
(non-invertible function like UNIX function)
- for the same data, the same hash



Remark: the hash $\varphi(\cdot)$ is assumed to be uninformative $\Rightarrow$ non-invertible.
(which is not true in information-theoretic sense).

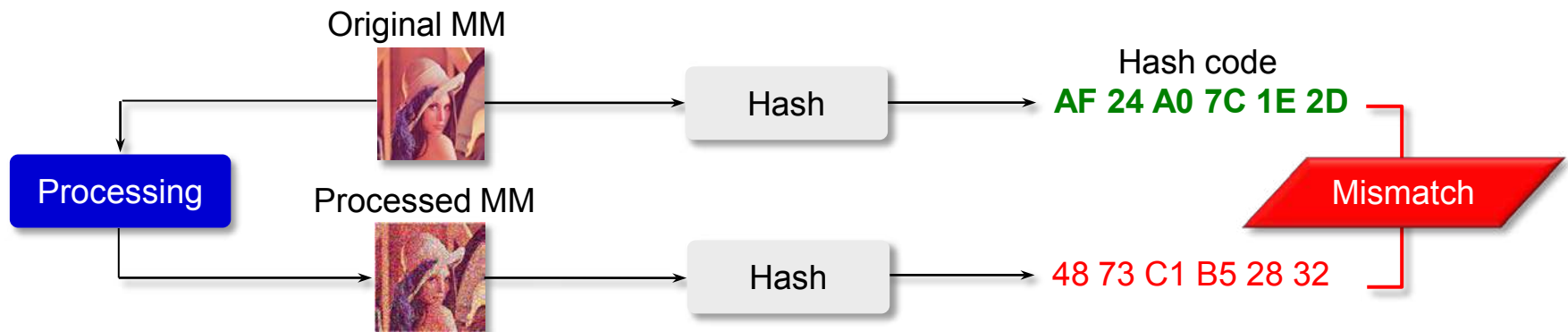
$$\underbrace{I(X; h(X))}_{\text{This is not true, since } h(X) \text{ is a function of } X!} = 0$$

This is not true, since $h(X)$ is a function of X !

Multimedia security: cryptographic solution

Why are traditional security tools not suitable for multimedia?

- Traditional *security*:
 - Cryptographic encryption (confidentiality of information)
 - Cryptographic hashes (authentication, trust, access control)
- Main concerns of classical crypto-based algorithms:
 - Sensitivity to noise and unintentional distortions in input data
 - Data handling in the encrypted domain



Multimedia privacy: data «degradation»

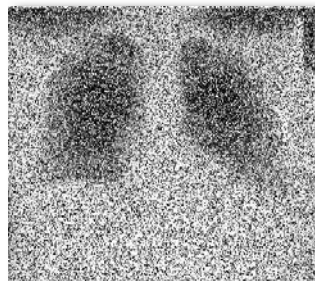
Why are traditional privacy tools not suitable for multimedia?

- Traditional *privacy protection* [1]:
 - Data owner (protect data): based on data degradation and randomization (noise addition, lossy compression, data removal, dim. reduction...)
 - Data user (protect requests): based on anonymization, randomized rules
- Main concerns of classical privacy preserving algorithms:
 - Reduction of accuracy (data utility)
 - Not very efficient against experienced attackers (sensitivity analysis)

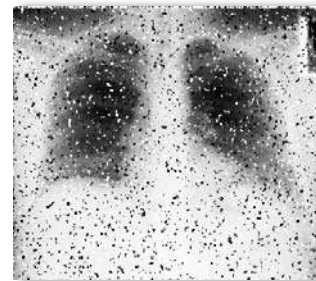
Original



50% removed based on key



Median 9x9



Adaptive recovery

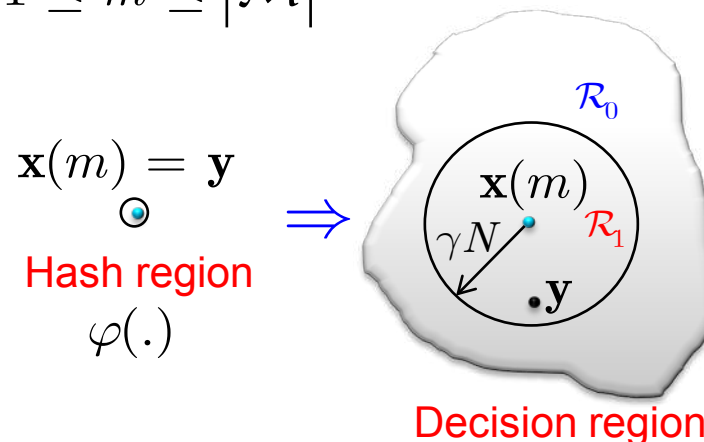


[1] C. Aggarwal and P. Yu, Privacy-Preserving Data Mining: Models and Algorithms, Springer, 2008.

Signal processing approach: authentication

- Use hypothesis testing
- Handle noisy data $\Rightarrow$ extended decision region

$$\begin{cases} H_0 : \mathbf{y} = \mathbf{x}' + \mathbf{z}, \\ H_1 : \mathbf{y} = \mathbf{x}(m) + \mathbf{z}, \end{cases}$$
$$\mathbf{x}' \neq \mathbf{x}(m), \text{ any } 1 \leq m \leq |\mathcal{M}|$$

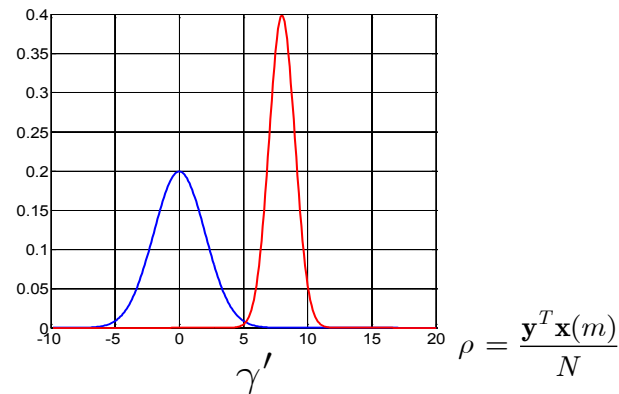


Gaussian assumption $\Rightarrow$ Euclidian distance $\equiv$ Sphere region

$$\mathcal{R}_1 : d^E(\mathbf{y}, \mathbf{x}(m)) = \|\mathbf{y} - \mathbf{x}(m)\|^2 \leq \gamma N \quad \Rightarrow \quad \mathcal{R}_1 : \mathbf{y}^T \mathbf{x}(m) \geq \gamma' N$$

Signal processing approach: authentication

$$\mathcal{R}_1 : \mathbf{y}^T \mathbf{x}(m) \geq \gamma' N$$



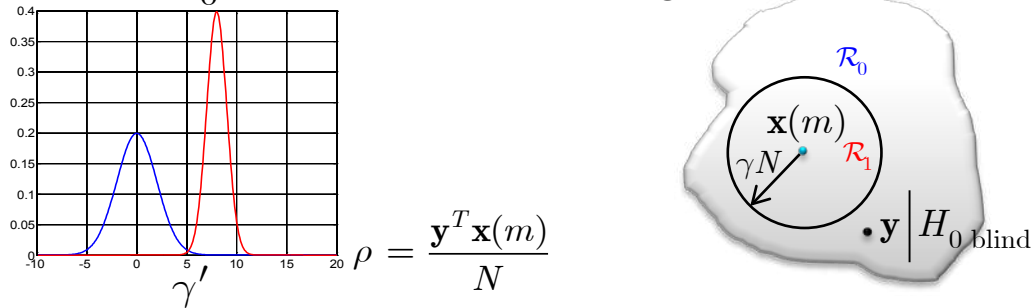
$$P_f = \Pr\left[\mathbf{y}^T \mathbf{x}(m) \geq \gamma' N \mid H_0\right] - \text{probability of false acceptance}$$

$$P_m = \Pr\left[\mathbf{y}^T \mathbf{x}(m) < \gamma' N \mid H_1\right] - \text{probability of miss}$$

Signal processing approach: authentication

Blind attacks

- $\mathbf{x}'$ under H_0 was assumed to be generated in blind way

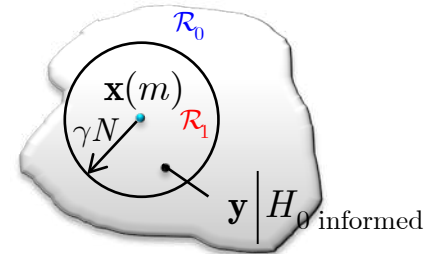


Informed attacks

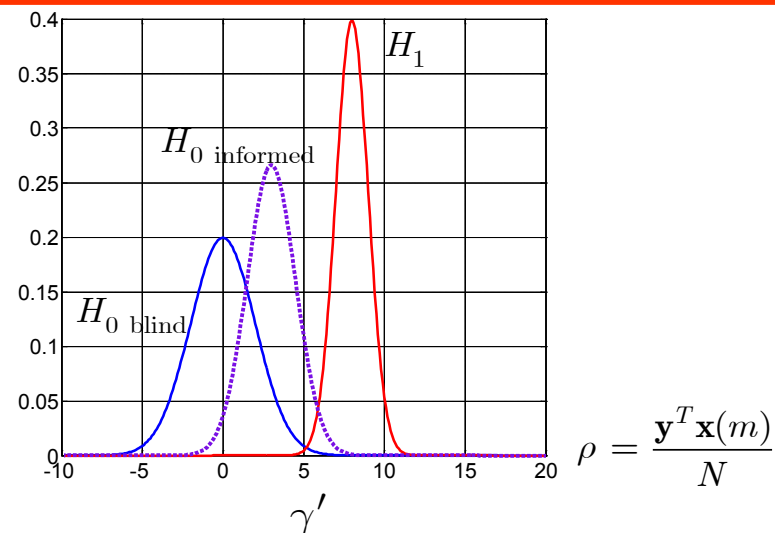
- Attacker can generate $\mathbf{x}'$ as close as possible to $\mathbf{x}(m)$, if it is disclosed!

$$P_f = \Pr[\mathbf{y}^T \mathbf{x}(m) \geq \gamma' N | H_0 \text{ blind}]$$

$$P_{sa} = \Pr[\mathbf{y}^T \mathbf{x}(m) \geq \gamma' N | H_0 \text{ informed}] - \text{prob. of succ. attack}$$



Signal processing approach: authentication

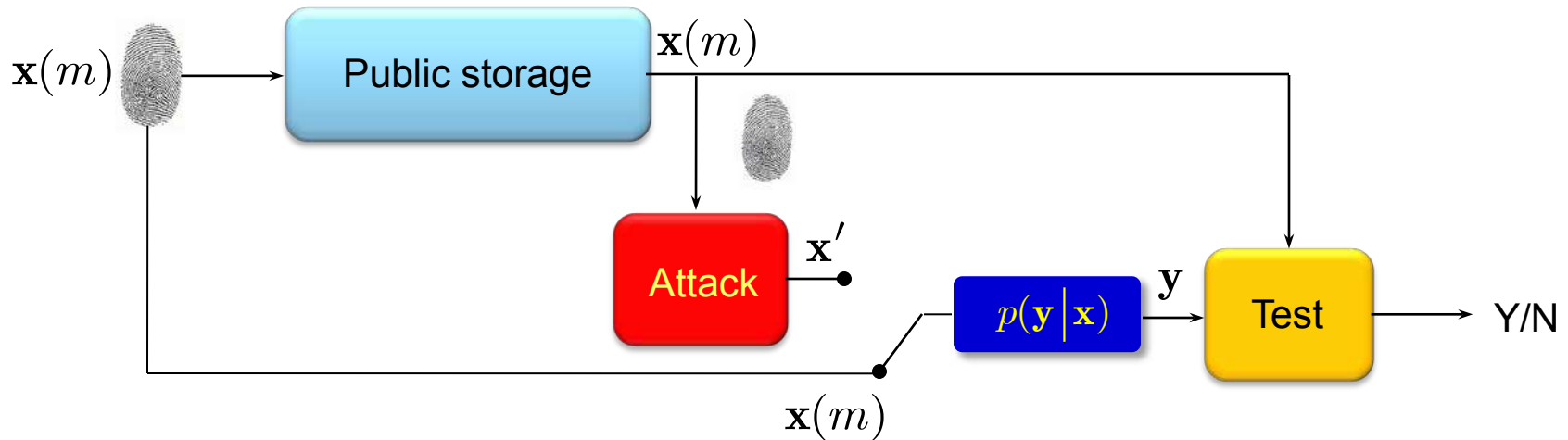


$$P_{sa} \geq P_f$$

Conclusion:

- the disclosure of $\mathbf{x}(m)$ is dangerous for the performance (security).

Signal processing approach: authentication



Open issues:

- $x(m)$ is stored in the public domain
 - no security
 - no privacy
- } $x(m)$ can be used for various attacks

Signal processing approach: identification

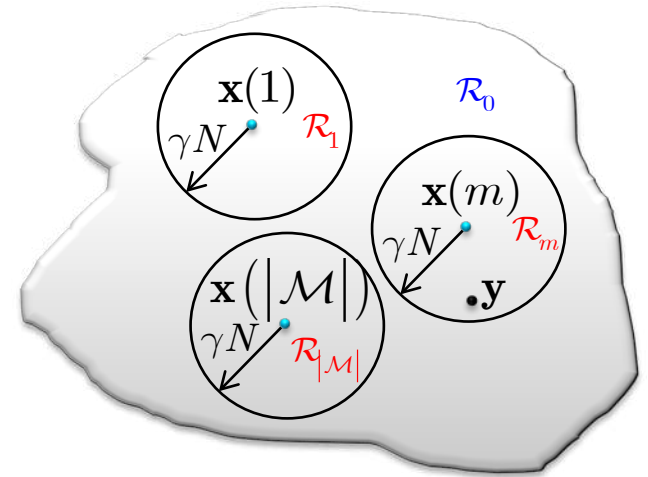
■ M-ary hypothesis testing

$$\begin{cases} H_0 : \mathbf{y} = \mathbf{x}' + \mathbf{z}, \\ H_1 : \mathbf{y} = \mathbf{x}(1) + \mathbf{z}, \\ \vdots \\ H_{|\mathcal{M}|} : \mathbf{y} = \mathbf{x}(|\mathcal{M}|) + \mathbf{z}, \\ \mathbf{x}' \neq \mathbf{x}(m), \text{ any } 1 \leq m \leq |\mathcal{M}| \end{cases}$$

$$P_f = \Pr \left[\bigcup_{m=1}^{|\mathcal{M}|} \mathbf{y}^T \mathbf{x}(m) \geq \gamma' N \mid H_{0 \text{ blind}} \right]$$

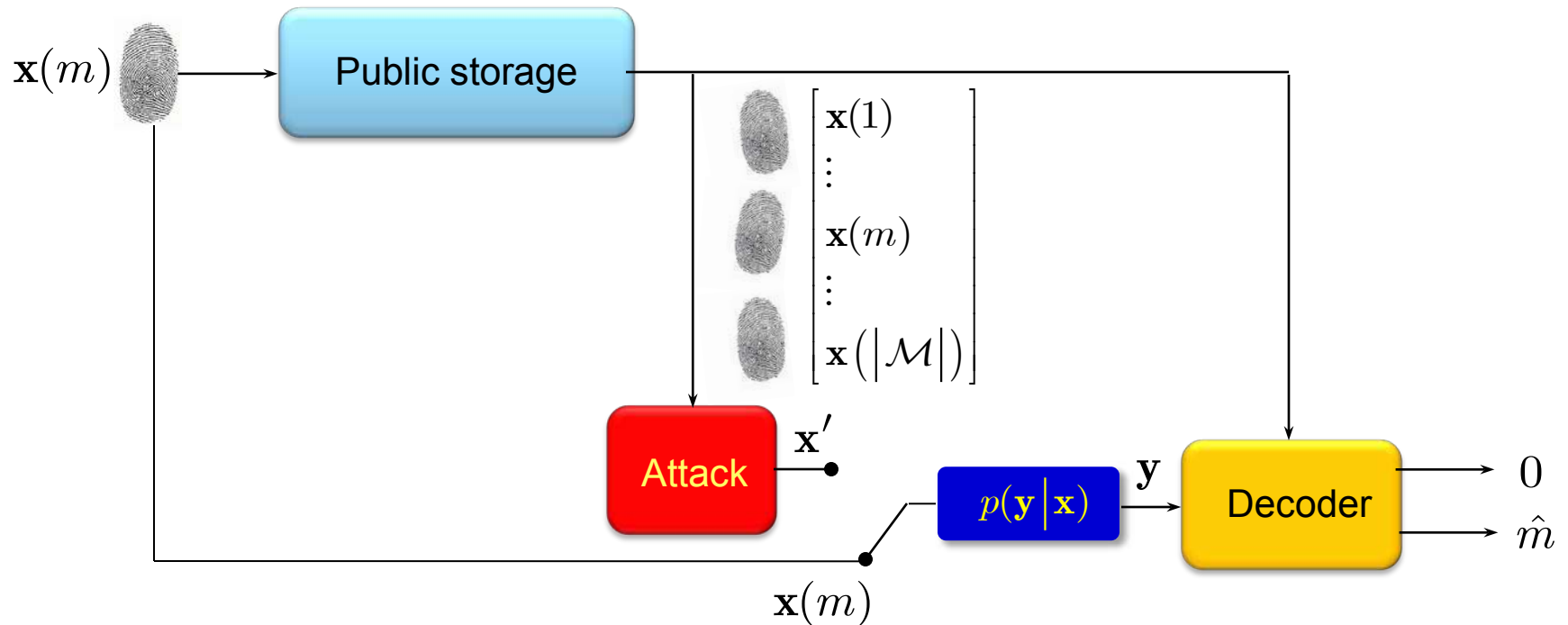
$$P_{sa} = \Pr \left[\bigcup_{m=1}^{|\mathcal{M}|} \mathbf{y}^T \mathbf{x}(m) \geq \gamma' N \mid H_{0 \text{ informed}} \right]$$

$$P_{ic} = \Pr \left[\underbrace{\mathbf{y}^T \mathbf{x}(m) < \gamma' N}_{\text{miss}} \bigcup \underbrace{\bigcup_{m' \neq m}^{|\mathcal{M}|} \mathbf{y}^T \mathbf{x}(m') \geq \gamma' N}_{\text{missacceptance}} \mid H_m \right]$$



$$\hat{m} = \arg \min_{1 \leq m \leq |\mathcal{M}|} \left(\|\mathbf{y} - \mathbf{x}(m)\|^2 \leq \gamma N \right)$$

Signal processing approach: identification

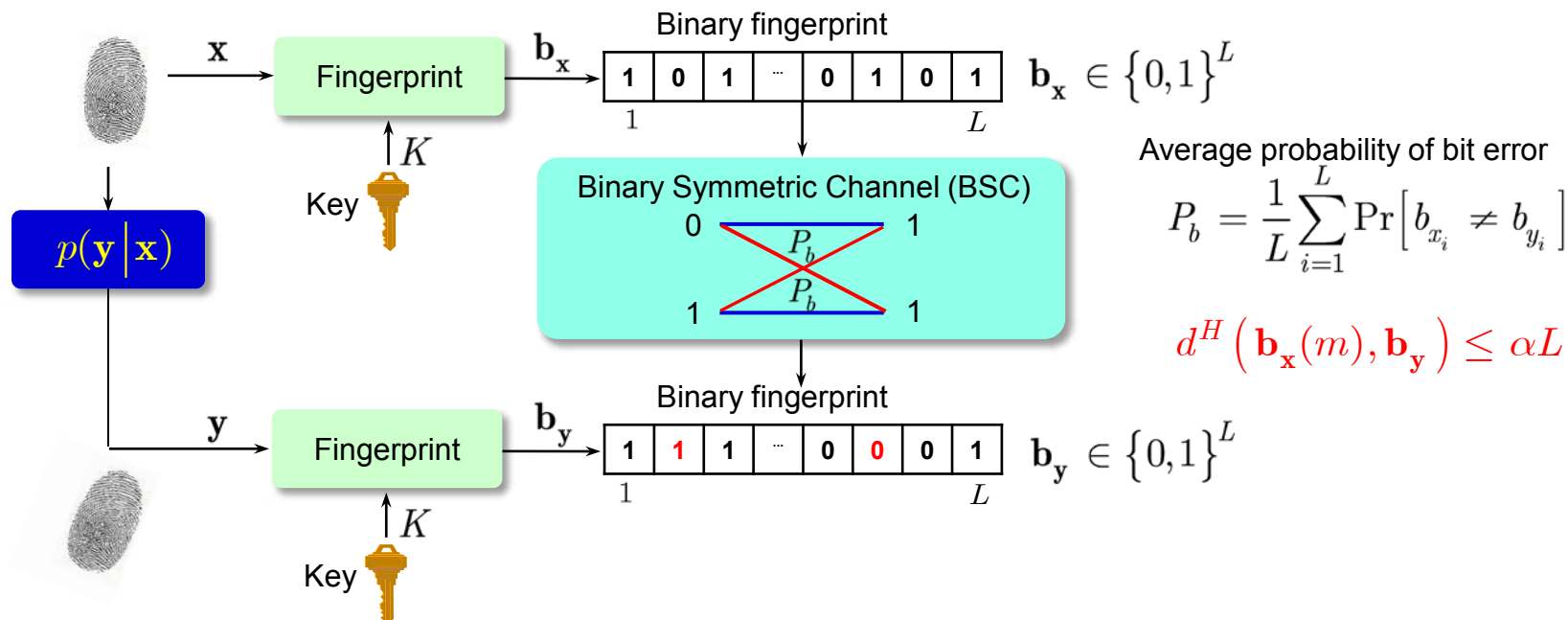


Open issues:

- all $\{\mathbf{x}(m)\}$ are stored in the public domain
 - no security
 - no privacy
- } $\mathbf{x}(m)$ can be used for various attacks

Signal processing approach: towards «robust hashing»

- If $\mathbf{x}(m)$ is open, the attacker can: (a) **deduce privacy** and (b) **design attacks**
- **Robust hashing**: an attempt to mimic crypto hashing but stay robust to signal processing (blur, noise, lossy compression, etc.) and geometric (affine, projective) distortions.



Signal processing approach: towards «robust hashing»

- Main assumptions:

- Robust hashing is robust to some minor modifications
- Robust hashing is non-invertible under

- (a) unknown key

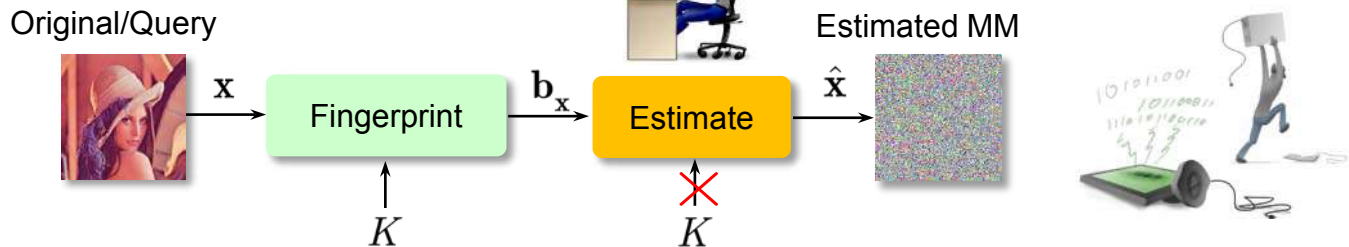
- (b) known key

such that $d^E(\mathbf{x}, \hat{\mathbf{x}}) \geq \gamma N$

This is possible due to loss of information

$$\mathbf{X} \rightarrow \mathbf{B}_x$$

$$\mathbf{Y} \rightarrow \mathbf{B}_y$$

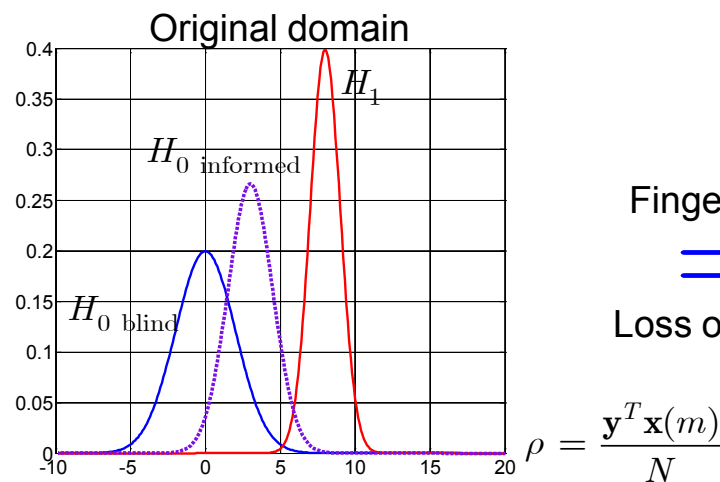


Remark: authentication/identification based on $(\mathbf{B}_x, \mathbf{B}_y)$ is less accurate than based on $(\mathbf{X}, \mathbf{Y})$

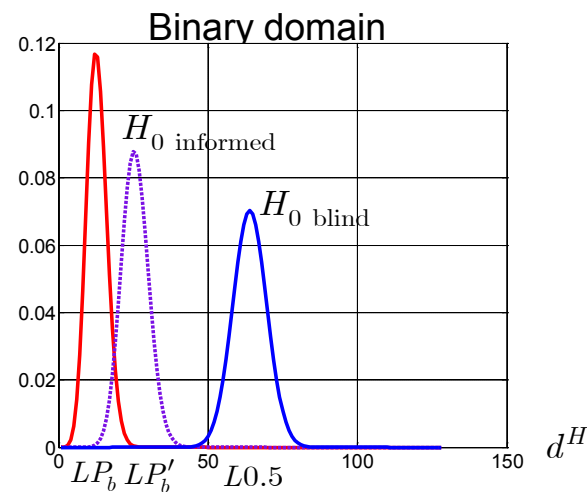
Signal processing approach: towards «robust hashing»

- Impact of reduction of dimensionality and binarization:

- FP: $\mathcal{R}^N \times \mathcal{K} \rightarrow \{0,1\}^L$

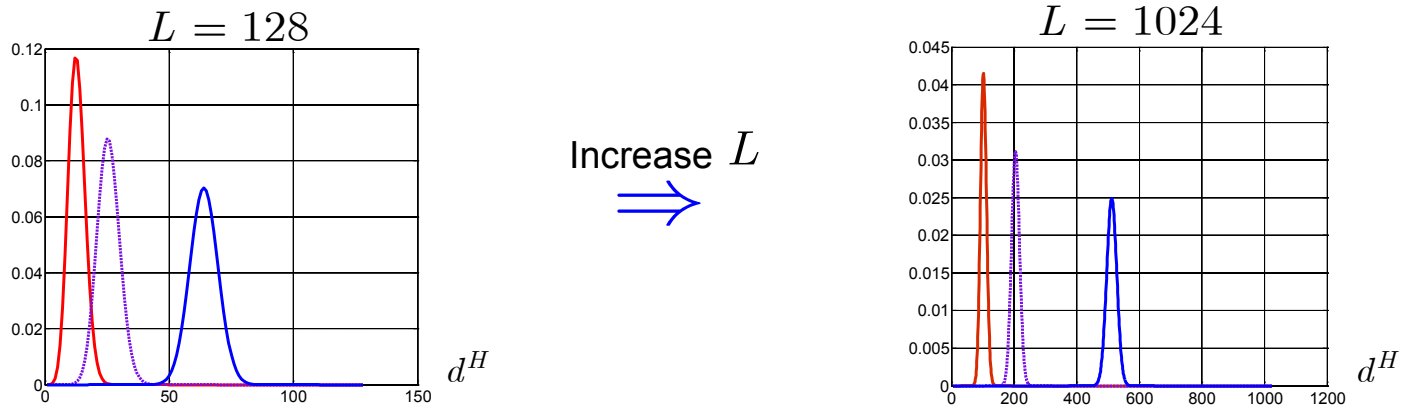


Fingerprinting
 $\Rightarrow$
 Loss of information



Hybrid solution

- Application of non-invertible transformations leads to the loss in performance
- One can compensate it by selecting longer L



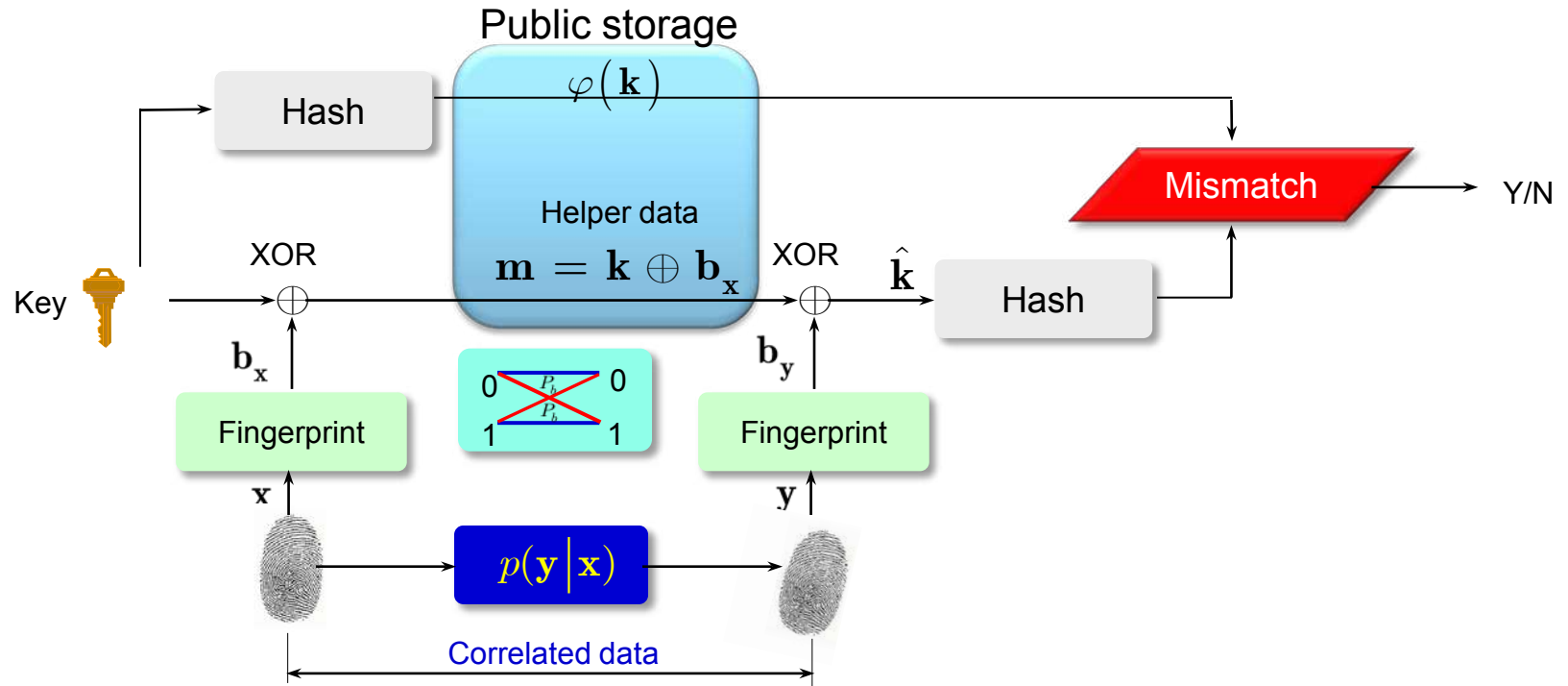
- In turn, longer L leads to better chances for image reconstruction!
- **Problem:** can one trade-off security/privacy – performance?

$\Rightarrow$ **Solution:** crypto + signal processing + coding

Hybrid solution: strategies

- **Several methods** (based on common idea):
 - fuzzy commitment
 - helper data
- Main strategy:
 - use additional helper data to correct errors
 - use a fact that $\mathbf{X}$ and $\mathbf{Y}$ are correlated as well as $\mathbf{B}_x$ and $\mathbf{B}_y$ for the hypothesis H_1 in authentication and the hypothesis H_m in identification

Hybrid solution: fuzzy commitment

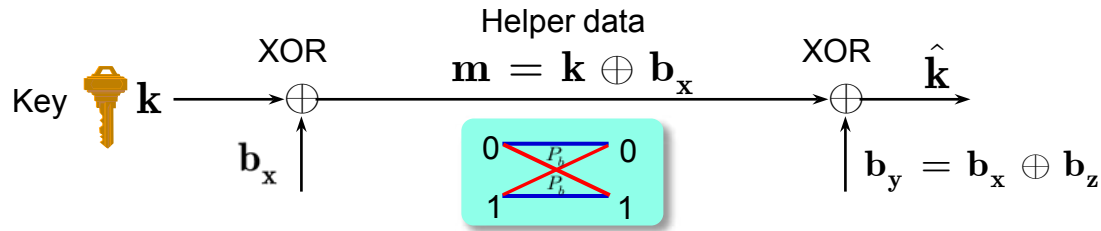


▪ Idea:

- k is used for every new session (i.i.d.) while b_x remains the same (non-renewable)
- b_x is used to lock the key (even, if it is not i.i.d.) $\Rightarrow$ analog of one-time-pad

Hybrid solution: fuzzy commitment

- Open issues:



- In general case,

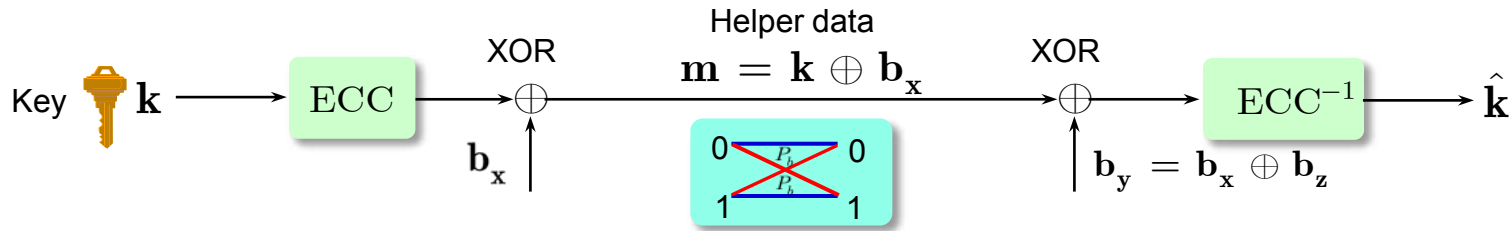
- $b_x \neq b_y \Rightarrow$ residual error

$$\hat{k} = k \oplus \underbrace{b_x \oplus b_y}_{b_z} = k \oplus b_z \Rightarrow \varphi(k) \neq \varphi(\hat{k})$$

- Idea:

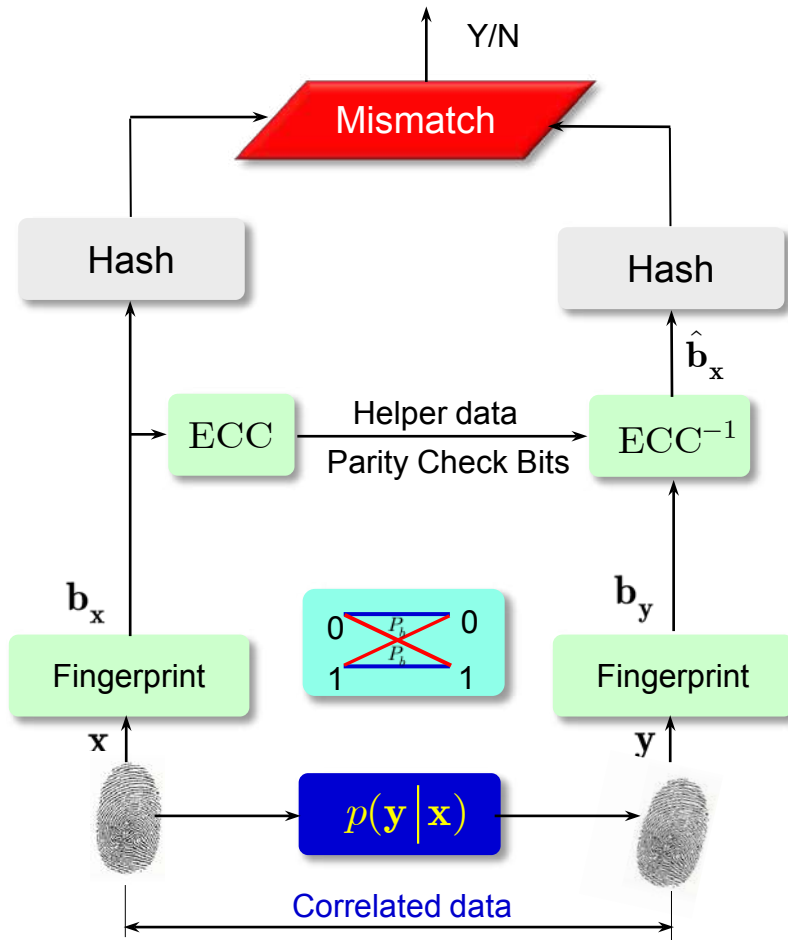
- add redundancy to k using error correction code to correct errors
- redundancy is proportional to $\approx LH_2(P_b)$

Hybrid solution: fuzzy commitment



- ECC^{-1} decoder corrects errors in $k \oplus b_z$
- Open issues:
 - $ECC(k)$ is not anymore i.i.d.
 - ⇒ smaller entropy: easier to predict and attack
 - In addition b_x is not i.i.d., the attacks are possible against both!

Hybrid solution: helper data based systems



- Open issues:
 - redundancy is proportional to $\approx LH_2(P_b)$
 - However, these methods represent the state-of-the-art!
 - It is up to you to find something better.